



LS ELECTRIC 천안 R-Center 구축설계

- 계산서 -
[기계]

2024.05

(주)가운종합건축사사무소

RTS-SAREK Load Calculation& Equipment Selection

RTS-SAREK Ver. C23

- 설계기준
- 냉난방 부하계산
- ☐ 공조기 선정
- Terminal Unit 선정
- ☐ 건물부하 분석
- ☐ Fan 선정

프로젝트 : LSELECTRIC(천안사업장) R-Center 구축
설 계 사 : 선진이앰씨

2024-05

RTS-SAREK



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■ 목 차

제 1장 부하계산

- 1-1. 설계개요
- 1-2. COOLING / HEATING LOAD SUMMARY
- 1-3. COOLING / HEATING LOAD CALCULATION

제 2장 공기조화설비

- 2-1. TERMINAL UNIT LOAD
- 2-2. EHP SELECTION

제 3장 환기설비

- 3-1. 환기량 선정기준
- 3-2. 풍량 선정
- 3-3. FAN 선정
- 3-3. 전열교환기 선정

제 4장 위생설비

- 4-1. 배수 설비

제 1장 부하계산

1-1. BUILDING DESIGN CONDITION

1-2. COOLING / HEATING LOAD SUMMARY

1-3. COOLING / HEATING LOAD CALCULATION

1-1. BUILDING DESIGN CONDITION

LSELECTRIC(천안사업장) R-Center 구축

2024-03-18

1. 설계 개요

1-1. 건물 개요/설계 기준

□ 건물 개요

Project Name :	LSELECTRIC(천안사업장) R-Center 구축											
Site Location :	충남 천안시 동남구 목천읍 삼성4길 56 부지 내						층 규 모 :					
건축 규모 / 설계사	대지면적 :	111,115 m²					건 축 주 :					
	건축면적 :	2,982 m²					건축설계 :	간운건축				
	연 면 적 :	3,562 m²					기계설계 :	선진이엠씨				

☐ Solar Data

Description	Time(h), 7/21												
	8	9	10	11	12	13	14	15	16	17	18	19	20
Apparent Solar time	7.4	8.4	9.4	10.4	11.4	12.4	13.4	14.4	15.4	16.4	17.4	18.4	19.4
Hour Angle	-69.6	-54.6	-39.6	-24.6	-9.6	5.4	20.4	35.4	50.4	65.4	80.4	95.4	110.4
Solar Altitude (β)	28.1	40.2	52.2	63.6	72.5	73.9	66.5	55.5	43.6	31.5	19.5	7.8	-3.3
Solar Azimuth (β)	-96.1	-87.5	-76.9	-61.1	-31.2	18.6	54.8	73.3	84.8	93.8	101.8	109.9	118.5
D. Normal Irr.(W/m²)	736.5	819.5	863.9	888	899.3	900.6	892.3	872.2	834.6	765.5	625.5	277.7	0
Standard Meridian :	-135			Longitude :			-127			Latitude :			36
Ground Reflection :	0.2			Clearness :			1			Month :			7
Solar Declination :	20.637			Equation of Time :			-6.3542, A = 1093, B = 0.186, C = 0.138						

□ 외기 온습도 기준

Atmospheric Pressure 100.248 kPa(90 m above Sea Level)

Description		Dry/Wet Bulb(℃ DB/WB)		D.R(℃)	Rel. Hum.(% RH)		Abs. Hum. (kg/kg')		Enthalpy(kJ/kg')	
A Contition	Cooling	32.5	25.8	10	59		0.0184		79.83	
	Heating	-12.1	-12.8		76		0.001		-9.69	

□ 인체 발열량 기준

(Unit : W/p)

Code	Type of Work	22 ℃		23 ℃		24 ℃		25 ℃		26 ℃		27 ℃		Remarks
		SH	LH	SH	LH	SH	LH	SH	LH	SH	LH	SH	LH	
A	경작업(사무소)	72	47	69	50	65	53	60	58	57	62	52	66	
B	경작업(은 행)	73	58	70	62	67	64	63	69	58	73	52	79	
C	앉은작업(식 당)	81	64	78	67	74	71	70	76	65	80	58	87	
D	정 좌(극 장)	66	27	63	30	60	33	58	35	56	37	51	42	
E	경작업(학 교)	71	34	67	37	64	41	59	45	56	49	51	53	
F	공장의 경작업	91	107	84	114	78	120	71	127	65	133	58	140	
G	댄스(댄스홀)	99	127	92	134	86	140	79	147	72	154	65	160	
H	공장의 중작업	115	149	108	156	100	164	94	170	134	177	79	185	
I	볼링(볼링장)	156	227	150	233	145	237	138	244	127	256	122	261	

□ Internal Load Data

Code	Service	People		Lighting		Equipment		Remarks	
		p/㎡	Radiant(%)	Code	W/㎡	Radiant(%)	W/㎡		Radiant(%)
CO	창고	0.1	58	A	10	67	0	0	
EQ	장비실	0.2	58	A	20	67	30	30	
KC	간이주방	0.3	58	A	20	67	20	30	
MA	편의시설	0.5	58	A	40	67	25	30	
ME	회의실	0.4	58	A	20	67	20	30	

OF	사무실	0.2	58	A	20	67	20	30
RO	로비/복도	0.2	58	A	20	67	0	0
TO	화장실/탈의실	0.1	58	A	20	67	0	0
WA	대기실	0.3	58	A	20	67	0	0

1-2. T.D & Glass/Wall/Roof/Partition Data

□Temperature Difference for Partitions

Code	Service	Temperature Difference (℃)		Remarks
		Cooling	Heating	
A	직접외기	5.1	31.3	
B	비공조실	2.55	15.65	
C	지중벽체	0	8.1	

□ Glass

Type	Service	K-value 'W/(㎡.K)'	AC	SHGC							Radiant	
				ID	0	40	50	60	70	80	He	(%)
G1	로이복층유리(G1)	1.384	1	25c	0.26	0.25	0.24	0.22	0.18	0.1	0.23	100
G3	도어(방풍실)	1.5	1	17b	0.6	0.59	0.57	0.51	0.4	0.21	0.53	100
G2	방풍문 유리	2.4	1	25c	0.26	0.25	0.24	0.22	0.18	0.1	0.23	100
G4	D2 유리도어로이	1.345	1	25c	0.26	0.25	0.24	0.22	0.18	0.1	0.23	100

P1 - F1(바닥) - 0.166

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (㎡.k)/W	K-Value W/(㎡.K)	Remarks
01	A02	내부저항-지붕/바닥			0.086		
02	C01	콘크리트(일반)	300	1.6	0.1875		
03	H11	압출법보온판(특호)	150	0.027	5.5556		
04	C01	콘크리트(일반)	60	1.6	0.0375		
05	A06	외부저항-벽체(간접)			0.15		
CTS Type :		Absorptance :		-		0.166	
Radiant :		Conductances :		-		K-Value : 0.166	

P2 - F2(연결통로바닥) - 0.131

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (㎡.k)/W	K-Value W/(㎡.K)	Remarks
01	A02	내부저항-지붕/바닥			0.086		
02	C01	콘크리트(일반)	150	1.6	0.0937		
03	H11	압출법보온판(특호)	200	0.027	7.4074		
04	A05	외부저항-바닥(직접)			0.043		
CTS Type :		Absorptance :		-		0.131	
Radiant :		Conductances :		-		K-Value : 0.131	

P3 - 간벽(임시) - 2.611

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (㎡.k)/W	K-Value W/(㎡.K)	Remarks
01	A01	내부저항-벽체			0.11		
02	M02	모르타르	18	1.51	0.0119		
03	C08	벽돌(일반)	180	1.29	0.1395		
04	M02	모르타르	18	1.51	0.0119		
05	A01	내부저항-벽체			0.11		
CTS Type :		Absorptance :		-		2.611	

Radiant :
Conductances :
K-Value :

-
-
2.611

R1 - 지붕 - 0.13

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (m².k)/W	K-Value W/(m².K)	Remarks
01	A05	외부저항-바닥(직접)			0.043		
02	C01	콘크리트(일반)	100	1.6	0.0625		
03	C01	콘크리트(일반)	150	1.6	0.0937		
04	H11	압출법보온판	200	0.027	7.4074		
05	A02	내부저항-지붕/바닥			0.086		
CTS Type :	32		Absorptance :	0.6		0.13	
Radiant :	60		Conductances :	17	K-Value :	0.13	

R2 - 지붕 - 0.141

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (m².k)/W	K-Value W/(m².K)	Remarks
01	A05	외부저항-바닥(직접)			0.043		
02	I08	미네랄울 보온판 1호 100K	50	0.034	1.4706		
03	I07	글라스울 네이처24K	186	0.034	5.4706		
04	A02	내부저항-지붕/바닥			0.086		
CTS Type :	32		Absorptance :	0.6		0.141	
Radiant :	60		Conductances :	17	K-Value :	0.141	

W1 - 외벽(직접) - 0.229

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (m².k)/W	K-Value W/(m².K)	Remarks
01	A01	내부저항-벽체			0.11		
02	G11	글라스울 64k	135	0.032	4.2188		
03	A03	외부저항-벽체(직접)			0.043		
CTS Type :	50		Absorptance :	0.6		0.229	
Radiant :	46		Conductances :	17	K-Value :	0.229	

W2 - 외벽(스팬드럴글라스) - 0.25

No	Code	Materials	THK mm	Conductivity W/(m.k)	Resistance (m².k)/W	K-Value W/(m².K)	Remarks
CTS Type :	50		Absorptance :	0.6		0.25	
Radiant :	46		Conductances :	17	K-Value :	0.25	

1-2. COOLING / HEATING LOAD SUMMARY

LSELECTRIC(천안사업장) R-Center 구축

2024-03-18

Description			Cooling Load (W) , 7/21							Heating Load
No.	Room Name	Q'ty	8	10	12	14	16	18	20	(W)
1001	1F_ AC. 대용량 모 터부하실 (방음)	1	10,997	22,643	25,175	26,603	27,186	27,231	8,127	15,487
1002	1F_ 부하 제어 및 기능시험실 (방음)	1	1,389	3,143	3,388	3,508	3,572	3,607	602	342
1003	1F_ B. 시험전원실	1	5,954	12,289	13,731	14,441	14,560	14,384	4,028	7,744
1004	1F_ I. PLC/Drive 검증시스템 및 신사 업 검증시스템	1	5,832	13,297	14,448	15,114	15,486	15,638	2,865	2,490
1005	1F_ AD. Drive 기능 시험실 (방음)	1	1,613	3,666	3,955	4,094	4,167	4,205	690	235
1006	1F_ A. Drive 시험 실	1	14,148	29,570	33,096	34,602	34,655	34,095	8,840	16,265
1007	1F_ G. 검증시스템 : 전기차/반도 체/AZS In-Line검 증시스템(160평)	1	27,145	61,297	66,170	68,663	70,140	71,111	12,879	9,550
1008	1F_F. 진동시험실	1	5,519	12,622	13,722	14,377	14,887	15,313	3,376	3,653
1009	1F_EA. 극한환경시 험실	1	1,113	2,543	2,758	2,879	2,952	2,997	578	643
1010	1F_ 제어룸	1	770	1,742	1,876	1,943	1,978	1,998	334	190
1011	1F_E. 환경시험실	1	8,687	19,786	21,414	22,290	22,798	23,094	4,240	3,935
1012	1F_C. 전기실&향온 항습실	1	1,635	3,725	4,031	4,196	4,293	4,347	799	743
1013	1F_C. EMI/EMS Shield실	1	2,762	6,289	6,805	7,082	7,245	7,338	1,345	1,243
1014	1F_CB. AMP실	1	983	2,255	2,454	2,574	2,647	2,694	554	742
1015	1F_CB. Control실	1	1,558	3,528	3,803	3,938	4,011	4,050	678	387
1016	1F_10M EMC CHAMBER	1	10,575	24,503	26,815	27,969	28,560	28,875	5,818	7,103
1017	1F_로비	1	8,592	13,517	13,029	12,766	12,554	12,136	4,808	14,783
1018	1F_방풍실	1	4,680	5,931	4,210	3,263	2,810	2,396	916	1,889
1019	1F_설비보전실	1	276	629	677	700	711	717	112	12
1020	1F_남자화장실	1	199	469	514	535	546	552	105	17
1021	1F_D. 전기적 기능 시험실	1	1,221	2,667	2,869	2,969	3,017	3,042	565	626
1022	1F_ ps	1	86	201	219	228	233	234	42	12
1023	1F_ eps/tps	1	72	164	176	183	186	187	29	7
1024	1F_ 고객 대기실-1	1	911	1,887	2,002	2,057	2,085	2,098	385	738
1025	1F_ 고객 대기실-2	1	794	1,770	1,885	1,940	1,968	1,981	268	21
1026	1F_ 고객 대기실-3	1	794	1,770	1,885	1,940	1,968	1,981	268	21
1027	1F_고객접견실	1	4,979	7,194	5,994	5,300	4,955	4,587	1,418	3,160
2001	2F_ Software Lab(20명)	1	2,336	5,335	5,773	5,990	6,104	6,165	1,073	405
2002	2F_중층 조망공간	1	383	889	971	1,012	1,035	1,047	206	116
2003	2F_여자화장실	1	131	312	345	363	372	377	85	54
2004	2F_남자화장실	1	227	535	589	615	630	639	136	83
2005	2F_ps	1	70	159	171	177	181	181	30	22
2006	2F_eps/tps	1	76	171	184	191	196	196	35	29
2007	2F_자료실	1	89	204	222	231	237	239	48	47
2008	2F_검증시스템 Control Room	1	1,179	2,371	2,537	2,619	2,662	2,687	644	1,615
2009	2F_세미나실/다목 적실(32석)	1	4,776	10,539	12,191	12,618	12,133	11,593	2,440	4,309
2010	2F_휴게공간	1	1,962	3,326	3,242	3,171	3,130	3,053	934	2,174
2011	2F_대회의실	1	1,751	3,777	4,019	4,138	4,200	4,233	681	768

2012	2F_OA/탕비실	1	2,235	3,037	2,361	1,979	1,797	1,616	540	1,297
2013	2F_사무실 (30석)	1	12,511	20,233	18,517	17,481	16,979	16,360	4,228	8,303
2014	2F_중회의실-1	1	688	1,526	1,622	1,671	1,696	1,710	238	71
2015	2F_중회의실-2	1	690	1,531	1,628	1,678	1,702	1,717	239	72
2016	2F_센터장실	1	1,947	2,755	2,208	1,899	1,756	1,619	455	1,142
2017	2F_복도	1	1,741	3,523	3,817	3,961	4,039	4,083	1,123	2,820
2018	2F_라운지	1	1,526	3,123	3,910	4,103	3,768	3,388	1,385	3,809
2019	2F_연결통로	1	20,818	28,829	24,104	25,687	34,024	35,508	15,642	19,822
Grand Total (kW)		46	178	351	366	376	387	387	95	139

1-3. COOLING / HEATING LOAD CALCULATION

LSELECTRIC(천안사업장) R-Center 구축2024-03-18

No.	Capacity				Cooling Load (W) , 7/21							Heating Load	
	Dire.	Tilt	K	m² (m³/h)	8	10	12	14	16	18	20	D.F	W
1001	1F_ AC. 대용량 모터부하실 (방음) 1 257 m², 5.5m				□ RTS : 4 (EQ) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50							-12.1/76 23/40	
G1	175	90	1.38	10.0	100	358	658	712	549	390	203	1.05	564
W1	175	90	0.23	173.6	-77	-	161	400	562	567	478	1.05	1,619
W1	265	90	0.23	117.3	-29	-	47	142	329	567	671	1.10	1,146
R2	0	90	0.14	257.0	109	101	106	126	157	192	224	1.20	1,786
P1	0/8.1		0.17	257.0	-	-	-	-	-	-	-	1.00	389
IF-S	C : 0.3ac/h			425	-	128	598	883	883	626	256		9,983
IF-L	H : 0.6ac/h,			849	2,693	2,764	2,764	2,764	2,764	2,764	2,764		
PE-S	0.20p/m², A,			52 p	1,072	2,636	2,948	3,100	3,175	3,213	728		
PE-L	(SH/LH :57.00/62.00W/p)				1,773	3,546	3,546	3,546	3,546	3,546	-		
EQ	30W/m²,			7,710 W	3,490	7,641	8,061	8,265	8,367	8,417	980		
LT	20W/m², 1.25,			6,425 W	2,137	5,505	6,286	6,665	6,854	6,949	1,823		
	Sensible Load				6,531	16,333	18,865	20,293	20,876	20,921	2,896		
	Total Load				10,997	22,643	25,175	26,603	27,186	27,231	8,127		15487
1002	1F_ 부하 제어 및 기능시험실 (방음) 1 40.5 m², 3m				□ RTS : 20 (EQ) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50							-12.1/76 23/40	
R2	0	90	0.14	40.5	17	16	17	20	25	30	35	1.20	281
P1	0/8.1		0.17	40.5	-	-	-	-	-	-	-	1.00	61
PE-S	0.20p/m², A,			9 p	184	453	507	533	546	553	126		
PE-L	(SH/LH :57.00/62.00W/p)				307	614	614	614	614	614	-		
EQ	30W/m²,			1,215 W	548	1,200	1,266	1,298	1,314	1,322	154		
LT	20W/m², 1.25,			1,013 W	333	860	984	1,043	1,073	1,088	287		
	Sensible Load				1,082	2,529	2,774	2,894	2,958	2,993	332		
	Total Load				1,389	3,143	3,388	3,508	3,572	3,607	602		342
1003	1F_ B.시험전원실 1 137.1 m², 5.5m				□ RTS : 4 (EQ) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50							-12.1/76 23/40	
G1	175	90	1.38	10.0	100	358	658	712	549	390	203	1.05	564
W1	175	90	0.23	74.2	-33	-	69	171	240	242	204	1.05	692
R2	0	90	0.14	137.1	58	54	57	67	84	103	120	1.20	953
P1	0/8.1		0.17	137.1	-	-	-	-	-	-	-	1.00	208
IF-S	C : 0.3ac/h			227	-	68	319	471	471	335	137		5,327
IF-L	H : 0.6ac/h,			453	1,439	1,476	1,476	1,476	1,476	1,476	1,476		
PE-S	0.20p/m², A,			28 p	577	1,420	1,588	1,669	1,710	1,730	392		
PE-L	(SH/LH :57.00/62.00W/p)				955	1,910	1,910	1,910	1,910	1,910	-		
EQ	30W/m²,			4,113 W	1,862	4,076	4,300	4,409	4,463	4,490	523		
LT	20W/m², 1.25,			3,428 W	1,140	2,937	3,354	3,556	3,657	3,708	973		
	Sensible Load				3,560	8,903	10,345	11,055	11,174	10,998	1,285		
	Total Load				5,954	12,289	13,731	14,441	14,560	14,384	4,028		7744
1004	1F_ I. PLC/Drive 검증시스템 및 신사업 검증시스템 1 175.2 m², 5.5m				□ RTS : 4 (EQ) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50							-12.1/76 23/40	
W1	175	90	0.23	108.0	-48	-	100	249	349	353	297	1.05	1,007
R2	0	90	0.14	175.2	74	69	72	86	107	131	153	1.20	1,218
P1	0/8.1		0.17	175.2	-	-	-	-	-	-	-	1.00	265
PE-S	0.20p/m², A,			36 p	742	1,825	2,041	2,146	2,198	2,224	504		
PE-L	(SH/LH :57.00/62.00W/p)				1,228	2,455	2,455	2,455	2,455	2,455	-		
EQ	30W/m²,			5,256 W	2,379	5,209	5,495	5,634	5,704	5,738	668		
LT	20W/m², 1.25,			4,380 W	1,457	3,753	4,285	4,544	4,673	4,737	1,243		
	Sensible Load				4,604	10,842	11,993	12,659	13,031	13,183	1,568		
	Total Load				5,832	13,297	14,448	15,114	15,486	15,638	2,865		2490
1005	1F_ AD. Drive 기능시험실 (방음) 1 48.6 m², 3m				□ RTS : 20 (EQ) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50							-12.1/76 23/40	
R2	0	90	0.14	23.2	10	9	10	11	14	17	20	1.20	161
P1	0/8.1		0.17	48.6	-	-	-	-	-	-	-	1.00	74
PE-S	0.20p/m², A,			10 p	204	503	563	592	607	614	140		
PE-L	(SH/LH :57.00/62.00W/p)				341	682	682	682	682	682	-		
EQ	30W/m²,			1,458 W	658	1,440	1,520	1,558	1,577	1,587	185		
LT	20W/m², 1.25,			1,215 W	400	1,032	1,180	1,251	1,287	1,305	345		
	Sensible Load				1,272	2,984	3,273	3,412	3,485	3,523	370		
	Total Load				1,613	3,666	3,955	4,094	4,167	4,205	690		235
1006	1F_ A. Drive 시험실 1 333.2 m², 4.5m				□ RTS : 5 (EQ) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50							-12.1/76 23/40	
G1	175	90	1.38	30.0	326	1,088	1,988	2,126	1,603	1,116	571	1.05	1,691
W1	175	90	0.23	153.6	-64	-	147	357	496	497	416	1.05	1,432

	W1	85	90	0.23	117.3	21	266	451	472	426	380	315	1.15	1,198
	R2	0	90	0.14	56.0	24	22	23	28	35	42	49	1.20	389
	P1	0/8.1		0.17	333.2	—	—	—	—	—	—	—	1.00	505
	P3	2.55/15.65		2.61	10.4	76	76	76	76	76	76	76	1.00	467
	IF-S	C : 0.3ac/h			450	—	136	633	935	935	663	271		10,583
	IF-L	H : 0.6ac/h,			900	2,852	2,927	2,927	2,927	2,927	2,927	2,927		
	PE-S	0.20p/㎡, A,			67 p	1,430	3,409	3,775	3,945	4,043	4,091	865		
	PE-L	(SH/LH :57.00/62.00W/p)				2,285	4,569	4,569	4,569	4,569	4,569	—		
	EQ	30W/㎡,			9,996 W	4,591	9,924	10,418	10,649	10,781	10,847	1,171		
	LT	20W/㎡, 1.25,			8,330 W	2,893	7,168	8,089	8,518	8,764	8,887	2,179		
		Sensible Load				9,011	22,074	25,600	27,106	27,159	26,599	3,228		
	Total Load					14,148	29,570	33,096	34,602	34,655	34,095	8,840		16265
1007	1F_G. 검증시스템 : 전기차/반도체/AZS In-Line검증 시스템(160평) 1 814.9 ㎡, 7.2m					□ RTS : 20 (EQ)					□ Out(℃/%) : 32.5/59			-12.1/76
						□ S.F : 10/10 %					□ In(℃/%) : 26/50			23/40
	W1	265	90	0.23	138.6	-30	—	57	168	386	666	788	1.10	1,354
	R2	0	90	0.14	814.9	346	321	338	400	499	609	710	1.20	5,663
	P1	0/8.1		0.17	814.9	—	—	—	—	—	—	—	1.00	1,234
	P3	2.55/15.65		2.61	28.8	212	212	212	212	212	212	212	1.00	1,299
	PE-S	0.20p/㎡, A,			163 p	3,332	8,205	9,183	9,657	9,894	10,013	2,282		
	PE-L	(SH/LH :57.00/62.00W/p)				5,558	11,117	11,117	11,117	11,117	11,117	—		
	EQ	30W/㎡,			24,447 W	11,026	24,149	25,480	26,125	26,448	26,609	3,106		
	LT	20W/㎡, 1.25,			20,373 W	6,701	17,305	19,783	20,984	21,584	21,885	5,781		
		Sensible Load				21,587	50,180	55,053	57,546	59,023	59,994	7,374		
	Total Load					27,145	61,297	66,170	68,663	70,140	71,111	12,879		9550
1008	1F_F. 진동시험실 1 166.7 ㎡, 5.5m					□ RTS : 4 (EQ)					□ Out(℃/%) : 32.5/59			-12.1/76
						□ S.F : 10/10 %					□ In(℃/%) : 26/50			23/40
	W1	265	90	0.23	117.3	-29	—	47	142	329	567	671	1.10	1,146
	W1	355	90	0.23	102.8	-32	—	52	122	187	233	265	1.20	1,095
	R2	0	90	0.14	166.7	70	65	69	82	102	125	146	1.20	1,159
	P1	0/8.1		0.17	166.7	—	—	—	—	—	—	—	1.00	253
	PE-S	0.20p/㎡, A,			34 p	701	1,724	1,928	2,027	2,076	2,101	476		
	PE-L	(SH/LH :57.00/62.00W/p)				1,159	2,319	2,319	2,319	2,319	2,319	—		
	EQ	30W/㎡,			5,001 W	2,264	4,956	5,229	5,361	5,427	5,460	635		
	LT	20W/㎡, 1.25,			4,168 W	1,386	3,571	4,078	4,324	4,447	4,508	1,183		
		Sensible Load				4,360	10,303	11,403	12,058	12,568	12,994	2,014		
	Total Load					5,519	12,622	13,722	14,377	14,887	15,313	3,376		3653
1009	1F_EA. 극한환경시험실 1 33.2 ㎡, 3m					□ RTS : 4 (EQ)					□ Out(℃/%) : 32.5/59			-12.1/76
						□ S.F : 10/10 %					□ In(℃/%) : 26/50			23/40
	W1	355	90	0.23	34.0	-11	—	17	40	62	77	88	1.20	362
	R2	0	90	0.14	33.2	14	13	14	16	20	25	29	1.20	231
	P1	0/8.1		0.17	33.2	—	—	—	—	—	—	—	1.00	50
	PE-S	0.20p/㎡, A,			7 p	144	355	397	417	427	433	98		
	PE-L	(SH/LH :57.00/62.00W/p)				239	477	477	477	477	477	—		
	EQ	30W/㎡,			996 W	451	987	1,041	1,068	1,081	1,087	127		
	LT	20W/㎡, 1.25,			830 W	276	711	812	861	885	898	236		
		Sensible Load				874	2,066	2,281	2,402	2,475	2,520	327		
	Total Load					1,113	2,543	2,758	2,879	2,952	2,997	578		643
1010	1F_ 제어룸 1 22.4 ㎡, 3m					□ RTS : 20 (EQ)					□ Out(℃/%) : 32.5/59			-12.1/76
						□ S.F : 10/10 %					□ In(℃/%) : 26/50			23/40
	R2	0	90	0.14	22.4	10	9	9	11	14	17	20	1.20	156
	P1	0/8.1		0.17	22.4	—	—	—	—	—	—	—	1.00	34
	PE-S	0.20p/㎡, A,			5 p	102	252	282	296	303	307	70		
	PE-L	(SH/LH :57.00/62.00W/p)				171	341	341	341	341	341	—		
	EQ	30W/㎡,			672 W	303	664	700	718	727	731	85		
	LT	20W/㎡, 1.25,			560 W	184	476	544	577	593	602	159		
		Sensible Load				599	1,401	1,535	1,602	1,637	1,657	184		
	Total Load					770	1,742	1,876	1,943	1,978	1,998	334		190
1011	1F_E. 환경시험실 1 261.6 ㎡, 5.5m					□ RTS : 4 (EQ)					□ Out(℃/%) : 32.5/59			-12.1/76
						□ S.F : 10/10 %					□ In(℃/%) : 26/50			23/40
	W1	355	90	0.23	161.5	-50	1	82	191	294	366	417	1.20	1,721
	R2	0	90	0.14	261.6	110	102	108	128	160	196	228	1.20	1,818
	P1	0/8.1		0.17	261.6	—	—	—	—	—	—	—	1.00	396
	PE-S	0.20p/㎡, A,			53 p	1,093	2,687	3,005	3,159	3,236	3,275	742		
	PE-L	(SH/LH :57.00/62.00W/p)				1,807	3,615	3,615	3,615	3,615	3,615	—		
	EQ	30W/㎡,			7,848 W	3,552	7,778	8,205	8,413	8,516	8,568	997		
	LT	20W/㎡, 1.25,			6,540 W	2,175	5,603	6,399	6,784	6,977	7,074	1,856		
		Sensible Load				6,880	16,171	17,799	18,675	19,183	19,479	2,352		
	Total Load					8,687	19,786	21,414	22,290	22,798	23,094	4,240		3935
1012	1F_C. 전기실&항온항습실 1 49.2 ㎡, 2.4m					□ RTS : 4 (EQ)					□ Out(℃/%) : 32.5/59			-12.1/76
						□ S.F : 10/10 %					□ In(℃/%) : 26/50			23/40
	W1	355	90	0.23	30.6	-10	—	16	36	56	69	79	1.20	326

R2	0	90	0.14	49.2	21	19	20	24	30	37	43	1.20	342
P1	0/8.1		0.17	49.2	—	—	—	—	—	—	—	1.00	75
PE-S	0.20p/m², A,			10 p	206	507	567	596	611	618	140		
PE-L	(SH/LH :57.00/62.00W/p)				341	682	682	682	682	682	—		
EQ	30W/m²,			1,476 W	668	1,463	1,543	1,582	1,602	1,611	188		
LT	20W/m², 1.25,			1,230 W	409	1,054	1,203	1,276	1,312	1,330	349		
	Sensible Load				1,294	3,043	3,349	3,514	3,611	3,665	443		
	Total Load				1,635	3,725	4,031	4,196	4,293	4,347	799		743
1013	1F_C. EMI/EMS Shield실				□ RTS : 4 (EQ)				□ Out(℃/%) : 32.5/59			-12.1/76	
	1 82.8 m², 2.4m				□ S.F : 10/10 %				□ In(℃/%) : 26/50			23/40	
W1	355	90	0.23	51.0	-16	—	26	60	93	116	132	1.20	543
R2	0	90	0.14	82.8	35	32	34	40	51	62	72	1.20	575
P1	0/8.1		0.17	82.8	—	—	—	—	—	—	—	1.00	125
PE-S	0.20p/m², A,			17 p	351	862	964	1,013	1,038	1,050	238		
PE-L	(SH/LH :57.00/62.00W/p)				580	1,159	1,159	1,159	1,159	1,159	—		
EQ	30W/m²,			2,484 W	1,124	2,462	2,597	2,663	2,696	2,712	316		
LT	20W/m², 1.25,			2,070 W	688	1,774	2,025	2,147	2,208	2,239	587		
	Sensible Load				2,182	5,130	5,646	5,923	6,086	6,179	746		
	Total Load				2,762	6,289	6,805	7,082	7,245	7,338	1,345		1243
1014	1F_CB. AMP실				□ RTS : 4 (EQ)				□ Out(℃/%) : 32.5/59			-12.1/76	
	1 29.9 m², 2.4m				□ S.F : 10/10 %				□ In(℃/%) : 26/50			23/40	
W1	355	90	0.23	45.9	-14	—	23	54	83	104	118	1.20	489
R2	0	90	0.14	29.9	13	12	12	15	18	22	26	1.20	208
P1	0/8.1		0.17	29.9	—	—	—	—	—	—	—	1.00	45
PE-S	0.20p/m², A,			6 p	124	304	340	358	366	371	84		
PE-L	(SH/LH :57.00/62.00W/p)				205	409	409	409	409	409	—		
EQ	30W/m²,			897 W	406	889	938	962	973	979	114		
LT	20W/m², 1.25,			748 W	249	641	732	776	798	809	212		
	Sensible Load				778	1,846	2,045	2,165	2,238	2,285	325		
	Total Load				983	2,255	2,454	2,574	2,647	2,694	554		742
1015	1F_CB. Control실				□ RTS : 20 (EQ)				□ Out(℃/%) : 32.5/59			-12.1/76	
	1 45.7 m², 2.4m				□ S.F : 10/10 %				□ In(℃/%) : 26/50			23/40	
R2	0	90	0.14	45.7	19	18	19	22	28	34	40	1.20	318
P1	0/8.1		0.17	45.7	—	—	—	—	—	—	—	1.00	69
PE-S	0.20p/m², A,			10 p	204	503	563	592	607	614	140		
PE-L	(SH/LH :57.00/62.00W/p)				341	682	682	682	682	682	—		
EQ	30W/m²,			1,371 W	618	1,354	1,429	1,465	1,483	1,492	174		
LT	20W/m², 1.25,			1,143 W	376	971	1,110	1,177	1,211	1,228	324		
	Sensible Load				1,217	2,846	3,121	3,256	3,329	3,368	373		
	Total Load				1,558	3,528	3,803	3,938	4,011	4,050	678		387
1016	1F_10M EMC CHAMBER				□ RTS : 20 (EQ)				□ Out(℃/%) : 32.5/59			-12.1/76	
	1 320.1 m², 11.4m				□ S.F : 10/10 %				□ In(℃/%) : 26/50			23/40	
W1	355	90	0.23	266.9	-79	4	136	315	483	602	685	1.20	2,844
W1	85	90	0.23	180.8	27	401	689	729	661	591	490	1.15	1,846
R1	0	90	0.13	320.1	118	109	115	136	170	207	242	1.20	1,928
P1	0/8.1		0.17	320.1	—	—	—	—	—	—	—	1.00	485
PE-S	0.20p/m², A,			65 p	1,329	3,272	3,662	3,851	3,945	3,993	910		
PE-L	(SH/LH :57.00/62.00W/p)				2,217	4,433	4,433	4,433	4,433	4,433	—		
EQ	30W/m²,			9,603 W	4,331	9,486	10,009	10,262	10,389	10,452	1,220		
LT	20W/m², 1.25,			8,003 W	2,632	6,798	7,771	8,243	8,479	8,597	2,271		
	Sensible Load				8,358	20,070	22,382	23,536	24,127	24,442	3,448		
	Total Load				10,575	24,503	26,815	27,969	28,560	28,875	5,818		7103
1017	1F_로비				□ RTS : 4 (RO)				□ Out(℃/%) : 32.5/59			-12.1/76	
	1 154.3 m², 4m				□ S.F : 10/10 %				□ In(℃/%) : 26/50			23/40	
G1	85	90	1.38	2.0	262	312	201	142	112	83	49	1.15	123
G2	85	90	2.40	24.0	3,084	3,719	2,463	1,816	1,479	1,126	674	1.15	2,558
W1	85	90	0.23	0.0		—	—	—	—	—	—	1.15	
W2	85	90	0.25	9.8	1	24	41	43	39	35	29	1.15	109
W1	85	90	0.23	0.0		—	—	—	—	—	—	1.15	
P1	0/8.1		0.17	154.3	—	—	—	—	—	—	—	1.00	234
P3	2.55/15.65		2.61	36.8	270	270	270	270	270	270	270	1.00	1,660
P3	2.55/15.65		2.61	127.2	935	935	935	935	935	935	935	1.00	5,737
IF-S	C : 0.3ac/h			186	—	56	262	386	386	274	112		4,362
IF-L	H : 0.6ac/h,			371	1,179	1,210	1,210	1,210	1,210	1,210	1,210		
PE-S	0.20p/m², A,			31 p	639	1,572	1,758	1,848	1,893	1,916	434		
PE-L	(SH/LH :57.00/62.00W/p)				1,057	2,114	2,114	2,114	2,114	2,114	—		
LT	20W/m², 1.25,			3,858 W	1,283	3,305	3,775	4,002	4,116	4,173	1,095		
	Sensible Load				6,356	10,193	9,705	9,442	9,230	8,812	2,313		
	Total Load				8,592	13,517	13,029	12,766	12,554	12,136	4,808		14783
1018	1F_방풍실				□ RTS : 6 (RO)				□ Out(℃/%) : 32.5/59			-12.1/76	
	1 19.5 m², 4m				□ S.F : 10/10 %				□ In(℃/%) : 26/50			23/40	
G1	85	90	1.38	6.0	829	969	612	416	325	241	118	1.15	370

G3	85	90	1.50	8.8	2,877	3,336	2,036	1,338	1,013	738	350	1.15	586
G1	85	90	1.38	3.2	442	517	326	222	173	129	63	1.15	197
W1	85	90	0.23	5.0	1	11	19	20	18	16	14	1.15	51
W1	85	90	0.23	0.0		—	—	—	—	—	—	1.15	
W1	85	90	0.23	10.0	1	23	39	41	37	33	27	1.15	102
P1	0/8.1		0.17	19.5	—	—	—	—	—	—	—	1.00	30
IF-S	C : 0.3ac/h			24	—	7	34	50	50	35	14		553
IF-L	H : 0.6ac/h,			47	152	156	156	156	156	156	156		
PE-S	0.20p/㎡, A,			4	86	208	230	239	244	247	50		
PE-L	(SH/LH :57.00/62.00W/p)				136	273	273	273	273	273	—		
LT	20W/㎡, 1.25,			488	171	431	485	508	521	528	124		
	Sensible Load				4,392	5,502	3,781	2,834	2,381	1,967	333		
	Total Load				4,680	5,931	4,210	3,263	2,810	2,396	916		1889
1019	1F_설비보전실					□ RTS : 20 (EQ)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 7.7 ㎡, 2.4m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	7.7	—	—	—	—	—	—	—	1.00	12
PE-S	0.20p/㎡, A,			2	41	101	113	118	121	123	28		
PE-L	(SH/LH :57.00/62.00W/p)				68	136	136	136	136	136	—		
EQ	30W/㎡,			231	104	228	241	247	250	251	29		
LT	20W/㎡, 1.25,			193	63	164	187	199	204	207	55		
	Sensible Load				208	493	541	564	575	581	58		
	Total Load				276	629	677	700	711	717	112		12
1020	1F_남자화장실					□ RTS : 20 (TO)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 10.9 ㎡, 2.4m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	10.9	—	—	—	—	—	—	—	1.00	17
PE-S	0.10p/㎡, A,			2	41	101	113	118	121	123	28		
PE-L	(SH/LH :57.00/62.00W/p)				68	136	136	136	136	136	—		
LT	20W/㎡, 1.25,			273	90	232	265	281	289	293	77		
	Sensible Load				131	333	378	399	410	416	55		
	Total Load				199	469	514	535	546	552	105		17
1021	1F_D. 전기적 기능시험실					□ RTS : 20 (EQ)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 34.3 ㎡, 2.4m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	34.3	—	—	—	—	—	—	—	1.00	52
P3	2.55/15.65		2.61	12.7	93	93	93	93	93	93	93	1.00	574
PE-S	0.20p/㎡, A,			7	143	352	394	415	425	430	98		
PE-L	(SH/LH :57.00/62.00W/p)				239	477	477	477	477	477	—		
EQ	30W/㎡,			1,029	464	1,016	1,072	1,100	1,113	1,120	131		
LT	20W/㎡, 1.25,			858	282	729	833	884	909	922	243		
	Sensible Load				982	2,190	2,392	2,492	2,540	2,565	338		
	Total Load				1,221	2,667	2,869	2,969	3,017	3,042	565		626
1022	1F_ ps					□ RTS : 20 (CO)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 7.8 ㎡, 3m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	7.8	—	—	—	—	—	—	—	1.00	12
PE-S	0.10p/㎡, A,			1	20	50	56	59	61	61	14		
PE-L	(SH/LH :57.00/62.00W/p)				34	68	68	68	68	68	—		
LT	10W/㎡, 1.25,			98	32	83	95	101	104	105	28		
	Sensible Load				52	133	151	160	165	166	21		
	Total Load				86	201	219	228	233	234	42		12
1023	1F_ eps/tps					□ RTS : 20 (CO)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 4.3 ㎡, 3m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	4.3	—	—	—	—	—	—	—	1.00	7
PE-S	0.10p/㎡, A,			1	20	50	56	59	61	61	14		
PE-L	(SH/LH :57.00/62.00W/p)				34	68	68	68	68	68	—		
LT	10W/㎡, 1.25,			54	18	46	52	56	57	58	15		
	Sensible Load				38	96	108	115	118	119	15		
	Total Load				72	164	176	183	186	187	29		7
1024	1F_ 고객 대기실-1					□ RTS : 20 (WA)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 14.1 ㎡, 3m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	14.1	—	—	—	—	—	—	—	1.00	21
P3	2.55/15.65		2.61	15.9	117	117	117	117	117	117	117	1.00	717
PE-S	0.55p/㎡, B,			8	166	410	459	482	494	500	114		
PE-L	(SH/LH :58.00/73.00W/p)				321	642	642	642	642	642	—		
EQ	30W/㎡,			423	191	418	441	452	458	460	54		
LT	20W/㎡, 1.25,			353	116	300	343	364	374	379	100		
	Sensible Load				590	1,245	1,360	1,415	1,443	1,456	256		
	Total Load				911	1,887	2,002	2,057	2,085	2,098	385		738
1025	1F_ 고객 대기실-2					□ RTS : 20 (WA)			□ Out(℃/%) : 32.5/59			-12.1/76	
	1 14.1 ㎡, 3m					□ S.F : 10/10 %			□ In(℃/%) : 26/50			23/40	
P1	0/8.1		0.17	14.1	—	—	—	—	—	—	—	1.00	21
PE-S	0.55p/㎡, B,			8	166	410	459	482	494	500	114		
PE-L	(SH/LH :58.00/73.00W/p)				321	642	642	642	642	642	—		

EQ LT	30W/㎡, 423 W				191	418	441	452	458	460	54		
	20W/㎡, 1.25, 353 W				116	300	343	364	374	379	100		
	Sensible Load				473	1,128	1,243	1,298	1,326	1,339	139		
	Total Load				794	1,770	1,885	1,940	1,968	1,981	268	21	
1026 1F_ 고객 대기실-3 1 14.1㎡, 3m					□ RTS : 20 (WA) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
P1	0/8.1	0.17	14.1		-	-	-	-	-	-	-	1.00	21
PE-S	0.55p/㎡, B,		8	p	166	410	459	482	494	500	114		
PE-L	(SH/LH :58.00/73.00W/p)				321	642	642	642	642	642	-		
EQ	30W/㎡,		423	W	191	418	441	452	458	460	54		
LT	20W/㎡, 1.25, 353 W				116	300	343	364	374	379	100		
	Sensible Load				473	1,128	1,243	1,298	1,326	1,339	139		
	Total Load				794	1,770	1,885	1,940	1,968	1,981	268	21	
1027 1F_고객접견실 1 45.8㎡, 4m					□ RTS : 6 (WA) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
G1	85	90	1.38	25.2	3,481	4,071	2,571	1,747	1,364	1,012	496	1.15	1,555
W1	85	90	0.23	23.8	4	54	92	97	87	78	64	1.15	243
P1	0/8.1		0.17	45.8	-	-	-	-	-	-	-	1.00	69
IF-S	C : 0.3ac/h			55	-	17	77	114	114	81	33	1,293	
IF-L	H : 0.6ac/h,			110	349	358	358	358	358	358	358		
PE-S	0.30p/㎡, A,			14	p	301	728	804	837	855	865	176	
PE-L	(SH/LH :57.00/62.00W/p)				477	955	955	955	955	955	-		
LT	20W/㎡, 1.25, 1,145 W				402	1,011	1,137	1,192	1,222	1,238	291		
	Sensible Load				4,153	5,881	4,681	3,987	3,642	3,274	477		
	Total Load				4,979	7,194	5,994	5,300	4,955	4,587	1,418	3160	
2001 2F_ Software Lab(20명) 1 67.2㎡, 3m					□ RTS : 20 (OF) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
R1	0	90	0.13	67.2	25	23	24	29	36	44	51	1.20	405
PE-S	0.30p/㎡, D,			21	p	422	1,039	1,162	1,222	1,252	1,267	289	
PE-L	(SH/LH :56.00/37.00W/p)				427	855	855	855	855	855	-		
EQ	30W/㎡,			2,016	W	909	1,991	2,101	2,154	2,181	2,194	256	
LT	20W/㎡, 1.25, 1,680 W				553	1,427	1,631	1,730	1,780	1,805	477		
	Sensible Load				1,909	4,480	4,918	5,135	5,249	5,310	585		
	Total Load				2,336	5,335	5,773	5,990	6,104	6,165	1,073	405	
2002 2F_중층 조망공간 1 19.2㎡, 5.6m					□ RTS : 20 (RO) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
R1	0	90	0.13	19.2	7	7	7	8	10	12	14	1.20	116
PE-S	0.20p/㎡, A,			4	p	82	201	225	237	243	246	56	
PE-L	(SH/LH :57.00/62.00W/p)				136	273	273	273	273	273	-		
LT	20W/㎡, 1.25, 480 W				158	408	466	494	509	516	136		
	Sensible Load				247	616	698	739	762	774	115		
	Total Load				383	889	971	1,012	1,035	1,047	206	116	
2003 2F_여자화장실 1 9㎡, 2.7m					□ RTS : 20 (TO) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
R1	0	90	0.13	9.0	3	3	3	4	5	6	7	1.20	54
PE-S	0.10p/㎡, A,			1	p	20	50	56	59	61	61	14	
PE-L	(SH/LH :57.00/62.00W/p)				34	68	68	68	68	68	-		
LT	20W/㎡, 1.25, 225 W				74	191	218	232	238	242	64		
	Sensible Load				97	244	277	295	304	309	47		
	Total Load				131	312	345	363	372	377	85	54	
2004 2F_남자화장실 1 13.8㎡, 2.7m					□ RTS : 20 (TO) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
R1	0	90	0.13	13.8	5	5	5	6	7	9	10	1.20	83
PE-S	0.10p/㎡, A,			2	p	41	101	113	118	121	123	28	
PE-L	(SH/LH :57.00/62.00W/p)				68	136	136	136	136	136	-		
LT	20W/㎡, 1.25, 345 W				113	293	335	355	366	371	98		
	Sensible Load				159	399	453	479	494	503	77		
	Total Load				227	535	589	615	630	639	136	83	
2005 2F_ps 1 3.7㎡, 2.7m					□ RTS : 20 (CO) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
R1	0	90	0.13	3.7	1	1	1	2	2	2	3	1.20	22
PE-S	0.10p/㎡, A,			1	p	20	50	56	59	61	61	14	
PE-L	(SH/LH :57.00/62.00W/p)				34	68	68	68	68	68	-		
LT	10W/㎡, 1.25, 47 W				15	40	46	48	50	50	13		
	Sensible Load				36	91	103	109	113	113	17		
	Total Load				70	159	171	177	181	181	30	22	
2006 2F_eps/tps 1 4.8㎡, 2.7m					□ RTS : 20 (CO) □ Out(℃/%) : 32.5/59 □ S.F : 10/10 % □ In(℃/%) : 26/50						-12.1/76 23/40		
R1	0	90	0.13	4.8	2	2	2	2	3	3	4	1.20	29
PE-S	0.10p/㎡, A,			1	p	20	50	56	59	61	61	14	

PE-L	(SH/LH :57.00/62.00W/p)				34	68	68	68	68	68	—		
LT	10W/㎡, 1.25, Sensible Load	60	W		20	51	58	62	64	64	17		
					42	103	116	123	128	128	20		
	Total Load				76	171	184	191	196	196	35		29
2007	2F_자료실				□ RTS : 20 (CO)		□ Out(℃/%) : 32.5/59		—12.1/76				
	1 7.8 ㎡, 3m				□ S.F : 10/10 %		□ In(℃/%) : 26/50		23/40				
R1	0	90	0.13	7.8	3	3	3	3	4	5	6	1.20	47
PE-S	0.10p/㎡, A,			1	20	50	56	59	61	61	14		
PE-L	(SH/LH :57.00/62.00W/p)				34	68	68	68	68	68	—		
LT	10W/㎡, 1.25, Sensible Load	98	W		32	83	95	101	104	105	28		
					55	136	154	163	169	171	27		
	Total Load				89	204	222	231	237	239	48		47
2008	2F_검증시스템 Control Room				□ RTS : 20 (EQ)		□ Out(℃/%) : 32.5/59		—12.1/76				
	1 27.8 ㎡, 3m				□ S.F : 10/10 %		□ In(℃/%) : 26/50		23/40				
R1	0	90	0.13	27.8	10	10	10	12	15	18	21	1.20	167
P3	2.55/15.65		2.61	32.1	236	236	236	236	236	236	236	1.00	1,448
PE-S	0.20p/㎡, A,			6	123	302	338	355	364	369	84		
PE-L	(SH/LH :57.00/62.00W/p)				205	409	409	409	409	409	—		
EQ	30W/㎡,		834	W	376	824	869	891	902	908	106		
LT	20W/㎡, 1.25, Sensible Load	695	W		229	590	675	716	736	747	197		
					974	1,962	2,128	2,210	2,253	2,278	459		
	Total Load				1,179	2,371	2,537	2,619	2,662	2,687	644		1615
2009	2F_세미나실/다목적실(32석)				□ RTS : 6 (EQ)		□ Out(℃/%) : 32.5/59		—12.1/76				
	1 77.6 ㎡, 3m				□ S.F : 10/10 %		□ In(℃/%) : 26/50		23/40				
G1	175	90	1.38	34.2	384	1,289	2,335	2,480	1,861	1,283	632	1.05	1,927
W1	175	90	0.23	2.7	—1	—	3	6	9	9	7	1.05	25
R1	0	90	0.13	77.6	28	26	28	33	41	51	59	1.20	467
P3	2.55/15.65		2.61	5.4	40	40	40	40	40	40	40	1.00	244
IF-S	C : 0.3ac/h			70	—	21	98	145	145	103	42		1,646
IF-L	H : 0.6ac/h,			140	444	455	455	455	455	455	455		
PE-S	0.45p/㎡, B,			35	767	1,851	2,045	2,129	2,175	2,201	447		
PE-L	(SH/LH :58.00/73.00W/p)				1,405	2,811	2,811	2,811	2,811	2,811	—		
EQ	30W/㎡,		2,328	W	1,073	2,334	2,449	2,499	2,526	2,542	265		
LT	20W/㎡, 1.25, Sensible Load	1,940	W		681	1,712	1,927	2,020	2,070	2,098	493		
					2,927	7,273	8,925	9,352	8,867	8,327	1,006		
	Total Load				4,776	10,539	12,191	12,618	12,133	11,593	2,440		4309
2010	2F_휴게공간				□ RTS : 6 (RO)		□ Out(℃/%) : 32.5/59		—12.1/76				
	1 44.1 ㎡, 3m				□ S.F : 10/10 %		□ In(℃/%) : 26/50		23/40				
G1	85	90	1.38	5.4	746	872	551	374	292	217	106	1.15	333
W1	85	90	0.23	13.0	2	30	51	53	48	43	35	1.15	133
R1	0	90	0.13	44.1	16	15	16	19	24	29	34	1.20	266
P3	2.55/15.65		2.61	11.1	82	82	82	82	82	82	82	1.00	501
IF-S	C : 0.3ac/h			40	—	12	56	83	83	59	24		941
IF-L	H : 0.6ac/h,			80	253	260	260	260	260	260	260		
PE-S	0.20p/㎡, A,			9	194	468	517	538	550	556	113		
PE-L	(SH/LH :57.00/62.00W/p)				307	614	614	614	614	614	—		
LT	20W/㎡, 1.25, Sensible Load	1,103	W		387	973	1,095	1,148	1,177	1,193	280		
					1,402	2,452	2,368	2,297	2,256	2,179	377		
	Total Load				1,962	3,326	3,242	3,171	3,130	3,053	934		2174
2011	2F_대회의실				□ RTS : 20 (RO)		□ Out(℃/%) : 32.5/59		—12.1/76				
	1 30.8 ㎡, 3m				□ S.F : 10/10 %		□ In(℃/%) : 26/50		23/40				
R1	0	90	0.13	30.8	11	11	11	13	16	20	23	1.20	186
P3	2.55/15.65		2.61	12.9	95	95	95	95	95	95	95	1.00	582
PE-S	0.50p/㎡, B,			16	333	819	917	965	988	1,000	228		
PE-L	(SH/LH :58.00/73.00W/p)				642	1,285	1,285	1,285	1,285	1,285	—		
EQ	30W/㎡,		924	W	417	913	963	987	1,000	1,006	117		
LT	20W/㎡, 1.25, Sensible Load	770	W		253	654	748	793	816	827	218		
					1,109	2,492	2,734	2,853	2,915	2,948	412		
	Total Load				1,751	3,777	4,019	4,138	4,200	4,233	681		768
2012	2F_OA/탕비실				□ RTS : 6 (RO)		□ Out(℃/%) : 32.5/59		—12.1/76				
	1 15.8 ㎡, 3m				□ S.F : 10/10 %		□ In(℃/%) : 26/50		23/40				
G1	85	90	1.38	12.9	1,782	2,084	1,316	894	698	518	254	1.15	796
W1	85	90	0.23	6.4	1	15	25	26	24	21	17	1.15	65
R1	0	90	0.13	15.8	6	5	6	7	8	10	12	1.20	95
IF-S	C : 0.3ac/h			15	—	5	21	31	31	22	9		341
IF-L	H : 0.6ac/h,			29	95	98	98	98	98	98	98		
PE-S	0.20p/㎡, A,			4	86	208	230	239	244	247	50		
PE-L	(SH/LH :57.00/62.00W/p)				136	273	273	273	273	273	—		
LT	20W/㎡, 1.25, Sensible Load	395	W		139	349	392	411	421	427	100		
					2,004	2,666	1,990	1,608	1,426	1,245	203		
	Total Load				2,235	3,037	2,361	1,979	1,797	1,616	540		1297

2013	2F_사무실 (30석)				□ RTS : 6 (OF)							□ Out(℃/%) : 32.5/59				-12.1/76		
	1 172.4 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40		
	G1	85	90	1.38	46.5	6,424	7,511	4,743	3,223	2,517	1,868	915	1.15	2,870				
	W1	85	90	0.23	23.2	3	53	91	95	85	76	63	1.15	237				
	R1	0	90	0.13	172.4	62	58	62	73	92	113	131	1.20	1,038				
	P3	2.55/15.65		2.61	11.1	82	82	82	82	82	82	82	1.00	501				
	IF-S	C : 0.3ac/h		156	-	47	219	324	324	230	94	3,657						
	IF-L	H : 0.6ac/h,		311	989	1,015	1,015	1,015	1,015	1,015	1,015							
	PE-S	0.20p/m², A,		35 p	754	1,819	2,010	2,093	2,137	2,163	439							
	PE-L	(SH/LH :57.00/62.00W/p)			1,194	2,387	2,387	2,387	2,387	2,387	-							
	EQ	20W/m²,		3,448 W	1,589	3,457	3,628	3,702	3,742	3,764	393							
	LT	20W/m², 1.25, Sensible Load		4,310 W	1,513	3,804	4,280	4,487	4,598	4,662	1,096							
	Total Load					12,511	20,233	18,517	17,481	16,979	16,360	4,228		8303				
2014	2F_중회의실-1				□ RTS : 20 (OF)							□ Out(℃/%) : 32.5/59				-12.1/76		
	1 11.8 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40		
	R1	0	90	0.13	11.8	4	4	4	5	6	8	9	1.20	71				
	PE-S	0.55p/m², B,		7 p	146	359	401	422	432	438	100							
	PE-L	(SH/LH :58.00/73.00W/p)			281	562	562	562	562	562	-							
	EQ	30W/m²,		354 W	160	350	369	378	383	385	45							
	LT	20W/m², 1.25, Sensible Load		295 W	97	251	286	304	313	317	84							
	Total Load					688	1,526	1,622	1,671	1,696	1,710	238		71				
	2015	2F_중회의실-2				□ RTS : 20 (OF)							□ Out(℃/%) : 32.5/59				-12.1/76	
		1 11.9 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40	
		R1	0	90	0.13	11.9	4	4	4	5	6	8	9	1.20	72			
		PE-S	0.55p/m², B,		7 p	146	359	401	422	432	438	100						
		PE-L	(SH/LH :58.00/73.00W/p)			281	562	562	562	562	562	-						
EQ		30W/m²,		357 W	161	353	372	382	386	389	45							
LT		20W/m², 1.25, Sensible Load		298 W	98	253	289	307	316	320	85							
Total Load					409	969	1,066	1,116	1,140	1,155	130							
Total Load					690	1,531	1,628	1,678	1,702	1,717	239		72					
2016		2F_센터장실				□ RTS : 6 (OF)							□ Out(℃/%) : 32.5/59				-12.1/76	
		1 14.8 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40	
		G1	85	90	1.38	10.5	1,451	1,696	1,071	728	568	422	207	1.15	648			
		W1	85	90	0.23	5.2	1	12	20	21	19	17	14	1.15	53			
	R1	0	90	0.13	14.8	5	5	5	6	8	10	11	1.20	89				
	P3	2.55/15.65		2.61	7.8	57	57	57	57	57	57	57	1.00	352				
	PE-S	0.20p/m², A,		3 p	65	156	172	179	183	185	38							
	PE-L	(SH/LH :57.00/62.00W/p)			102	205	205	205	205	205	-							
	EQ	20W/m²,		296 W	136	297	311	318	321	323	34							
	LT	20W/m², 1.25, Sensible Load		370 W	130	327	367	385	395	400	94							
	Total Load					1,845	2,550	2,003	1,694	1,551	1,414	247						
	Total Load					1,947	2,755	2,208	1,899	1,756	1,619	455		1142				
	2017	2F_복도				□ RTS : 20 (RO)							□ Out(℃/%) : 32.5/59				-12.1/76	
1 68.2 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40			
R1		0	90	0.13	68.2	25	23	25	29	36	44	51	1.20	411				
P3		2.55/15.65		2.61	53.4	392	392	392	392	392	392	392	1.00	2,409				
PE-S		0.20p/m², A,		14 p	286	705	789	829	850	860	196							
PE-L		(SH/LH :57.00/62.00W/p)			477	955	955	955	955	955	-							
LT		20W/m², 1.25, Sensible Load		1,705 W	561	1,448	1,656	1,756	1,806	1,832	484							
Total Load					1,264	2,568	2,862	3,006	3,084	3,128	800							
Total Load					1,741	3,523	3,817	3,961	4,039	4,083	1,123		2820					
2018		2F_라운지				□ RTS : 4 (RO)							□ Out(℃/%) : 32.5/59				-12.1/76	
		1 34.9 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40	
		G4	85	90	1.34	2.0	260	309	199	140	110	82	48	1.15	119			
		G1	175	90	1.38	22.2	223	794	1,461	1,580	1,218	867	451	1.05	1,251			
	W1	85	90	0.23	13.3	2	30	51	54	49	44	36	1.15	136				
	W1	175	90	0.23	24.6	-11	-	23	57	80	80	68	1.05	229				
	R1	0	90	0.13	34.9	13	12	12	15	19	23	26	1.20	210				
	P3	2.55/15.65		2.61	24.9	183	183	183	183	183	183	183	1.00	1,123				
	IF-S	C : 0.3ac/h		32	-	10	45	66	66	47	19	741						
	IF-L	H : 0.6ac/h,		63	203	208	208	208	208	208	208							
	PE-S	0.20p/m², A,		7 p	144	355	397	417	427	433	98							
	PE-L	(SH/LH :57.00/62.00W/p)			239	477	477	477	477	477	-							
	LT	20W/m², 1.25, Sensible Load		873 W	290	748	854	906	931	944	248							
Total Load					1,084	2,438	3,225	3,418	3,083	2,703	704							
Total Load					1,526	3,123	3,910	4,103	3,768	3,388	1,385		3809					
2019	2F_연결통로				□ RTS : 4 (RO)							□ Out(℃/%) : 32.5/59				-12.1/76		
	1 116 m², 3m				□ S.F : 10/10 %							□ In(℃/%) : 26/50				23/40		
G1	85	90	1.38	126.5	16,593	19,703	12,724	8,991	7,058	5,252	3,092	1.15	7,804					

G1	265	90	1.38	126.5	1,214	2,859	4,420	9,379	19,470	22,686	10,144	1.10	7,465
W1	265	90	0.23	36.5	−9	−	15	44	102	177	209	1.10	357
W1	85	90	0.23	36.5	5	81	140	148	134	120	100	1.15	373
R2	0	90	0.14	116.0	49	45	48	57	71	87	101	1.20	806
P2	5.1/31.3		0.13	116.0	91	91	91	91	91	91	91	1.00	559
IF−S	C : 0.3ac/h			105	−	32	148	218	218	155	63		2,458
IF−L	H : 0.6ac/h,			209	665	683	683	683	683	683	683		
PE−S	0.20p/m², A,	24	p		495	1,217	1,361	1,431	1,466	1,483	336		
PE−L	(SH/LH :57.00/62.00W/p)				818	1,637	1,637	1,637	1,637	1,637	−		
LT	20W/m², 1.25,	2,900	W		964	2,485	2,837	3,008	3,094	3,137	823		
	Sensible Load				19,335	26,509	21,784	23,367	31,704	33,188	7,223		
	Total Load				20,818	28,829	24,104	25,687	34,024	35,508	15,642		19822

제 2장 공기조화설비

2-1 TERMINAL UNIT LOAD

2-2. EHP SELECTION

2-1. TERMINAL UNIT LOAD

LSELECTRIC(천안사업장) R-Center 구축

2024-03-18

RM No.	Service	Qty	Area		Room Peak Cooling Load			Zone Peak Cooling Load			Heating	Appl.	Data
			m²	(W)			(W)			W	Cool	Heat	
				hr	SH	LH	RA	SH	LH				RA
EHP1	Others		189	—	14,651	2,660	14,651	2,660	3,843	—	—		
진동시험실							□ Peak time 18 h						
1008	1F_F. 진동시험실	1	167	18 : 00	12,994	2,319	12,994	2,319	3,653	WGRPFI	WGRPFP		
1010	1F_ 제어룸	1	22	18 : 00	1,657	341	1,657	341	190	WGRPFI	WGRPFP		
EHP2	Others		262	—	19,479	3,615	19,479	3,615	3,935	—	—		
환경시험실							□ Peak time 18 h						
1011	1F_E. 환경시험실	1	262	18 : 00	19,479	3,615	19,479	3,615	3,935	WGRPFI	WGRPFP		
EHP3	Others		298	—	23,945	6,924	23,928	6,924	15,829	—	—		
대용량 모터부하실							□ Peak time 17 h						
1001	1F_AC. 대용량 모 터부하실 (방음)	1	257	17 : 00	20,952	6,310	20,952	6,310	15,487	WGRPFI	WGRPFP		
1002	1F_ 부하 제어 및 기 능시험실 (방음)	1	40	18 : 00	2,993	614	2,977	614	342	WGRPFI	WGRPFP		
EHP4	Others		137	—	11,175	3,386	11,175	3,386	7,744	—	—		
시험전원실							□ Peak time 15 h						
1003	1F_B.시험전원실	1	137	15 : 00	11,175	3,386	11,175	3,386	7,744	WGRPFI	WGRPFP		
EHP5	Others		333	—	27,269	7,496	27,269	7,496	16,265	—	—		
Drive 시험실							□ Peak time 15 h						
1006	1F_A. Drive 시험실	1	333	15 : 00	27,269	7,496	27,269	7,496	16,265	WGRPFI	WGRPFP		
EHP6	Others		325	—	26,296	7,722	24,791	7,722	19,584	—	—		
1층사무공간							□ Peak time 11 h						
1005	1F_AD. Drive 기능 시험실 (방음)	1	49	18 : 00	3,523	682	3,159	682	235	WGRPFI	WGRPFP		
1021	1F_D. 전기적 기능 시험실	1	34	18 : 00	2,565	477	2,313	477	626	WGRPFI	WGRPFP		
1024	1F_ 고객 대기실-1	1	14	18 : 00	1,456	642	1,314	642	738	WGRPFI	WGRPFP		
1025	1F_ 고객 대기실-2	1	14	18 : 00	1,339	642	1,197	642	21	WGRPFI	WGRPFP		
1026	1F_ 고객 대기실-3	1	14	18 : 00	1,339	642	1,197	642	21	WGRPFI	WGRPFP		
1027	1F_고객접견실	1	46	10 : 00	5,881	1,313	5,450	1,313	3,160	WGRPFI	WGRPFP		
1017	1F_로비	1	154	10 : 00	10,193	3,324	10,162	3,324	14,783	WGRPFI	WGRPFP		
EHP7	Others		678	—	84,583	15,380	78,406	15,380	45,310	—	—		
2층사무공간							□ Peak time 17 h						
2001	2F_ Software Lab(20명)	1	67	18 : 00	5,310	855	5,280	855	405	WGRPFI	WGRPFP		
2008	2F_검증시스템 Control Room	1	28	18 : 00	2,278	409	2,264	409	1,615	WGRPFI	WGRPFP		
2009	2F_세미나실/다목적 실(32석)	1	78	14 : 00	9,352	3,266	8,612	3,266	4,309	WGRPFI	WGRPFP		
2010	2F_휴게공간	1	44	11 : 00	2,461	874	2,222	874	2,174	WGRPFI	WGRPFP		
2011	2F_대회의실	1	31	18 : 00	2,948	1,285	2,931	1,285	768	WGRPFI	WGRPFP		
2013	2F_사무실 (30석)	1	172	10 : 00	16,831	3,402	13,301	3,402	8,303	WGRPFI	WGRPFP		
2014	2F_중회의실-1	1	12	18 : 00	1,148	562	1,141	562	71	WGRPFI	WGRPFP		
2015	2F_중회의실-2	1	12	18 : 00	1,155	562	1,147	562	72	WGRPFI	WGRPFP		
2016	2F_센터장실	1	15	10 : 00	2,550	205	1,487	205	1,142	WGRPFI	WGRPFP		
2017	2F_복도	1	68	18 : 00	3,128	955	3,106	955	2,820	WGRPFI	WGRPFP		
2018	2F_라운지	1	35	14 : 00	3,418	685	2,907	685	3,809	WGRPFI	WGRPFP		
2019	2F_연결통로	1	116	17 : 00	34,007	2,320	34,007	2,320	19,822	WGRPFI	WGRPFP		

2-1. EHP SELECTION

LSELECTRIC(천안사업장) R-Center 구축

2024-03-18

Room No.	형 식	수량	용 량(W)			면적	장비 선정	개수	형식	
			냉방 부하							난방부하
			현열	잠열	전열					
1008	1F_F. 진동시험실	1	12,994	2,319	15,313	3653	166.7	29.0	2	스탠다드형
1010	1F_ 제어룸	1	1,657	341	1,998	190	22.4	5.2	1	4WAY
1011	1F_E. 환경시험실	1	19,479	3,615	23,094	3935	261.6	29.0	3	스탠다드형
1001	1F_ AC. 대용량 모터부하 실 (방음)	1	20,952	6,310	27,262	15487	257	29.0	3	스탠다드형
1002	1F_ 부하 제어 및 기능시 험실 (방음)	1	2,993	614	3,607	342	40.5	10.0	1	4WAY
1003	1F_ B.시험전원실	1	11,175	3,386	14,561	7744	137.1	29.0	3	스탠다드형
1006	1F_ A. Drive 시험실	1	27,269	7,496	34,765	16265	333.2	29.0	4	스탠다드형
1005	1F_ AD. Drive 기능시험 실 (방음)	1	3,523	682	4,205	235	48.6	13.0	1	4WAY
1021	1F_D. 전기적 기능시험실	1	2,565	477	3,042	626	34.3	7.2	1	4WAY
1024	1F_ 고객 대기실-1	1	1,456	642	2,098	738	14.1	2.3	1	1WAY
1025	1F_ 고객 대기실-2	1	1,339	642	1,981	21	14.1	2.3	1	1WAY
1026	1F_ 고객 대기실-3	1	1,339	642	1,981	21	14.1	2.3	1	1WAY
1027	1F_고객접견실	1	5,881	1,313	7,194	3160	45.8	10.0	1	4WAY
1017	1F_로비	1	10,193	3,324	13,517	14783	154.3	7.2	4	4WAY
2001	2F_ Software Lab(20명)	1	5,310	855	6,165	405	67.2	7.2	2	4WAY
2008	2F_검증시스템 Control Room	1	2,278	409	2,687	1615	27.8	7.2	1	4WAY
2009	2F_세미나실/다목적실 (32석)	1	9,352	3,266	12,618	4309	77.6	10.0	2	4WAY
2010	2F_휴게공간	1	2,461	874	3,335	2174	44.1	8.3	1	4WAY
2011	2F_대회의실	1	2,948	1,285	4,233	768	30.8	6.0	1	4WAY
2013	2F_사무실 (30석)	1	16,831	3,402	20,233	8303	172.4	8.3	4	4WAY
2014	2F_중회의실-1	1	1,148	562	1,710	71	11.8	2.3	1	1WAY
2015	2F_중회의실-2	1	1,155	562	1,717	72	11.9	2.3	1	1WAY
2016	2F_센터장실	1	2,550	205	2,755	1142	14.8	3.2	1	1WAY
2017	2F_복도	1	3,128	955	4,083	2820	68.2	5.2	2	2WAY
2018	2F_라운지	1	3,418	685	4,103	3809	34.9	5.2	1	2WAY
2019	2F_연결통로	1	34,007	2,320	36,327	19822	116	7.2	3	4WAY

제 3장 환기 설비

3-1. 환기량 선정기준

3-2. 풍량 선정

3-3. FAN 선정

3-4. 전열교환기 선정

3-1. 실용도별 환기량 선정기준

구 분	환기량	비 고
화장실	10~15 회/hr	
기계실,PIT	5 회/hr	
보일러실	10 회/hr	
발전기실	10 회/hr	
전기실	$Q = \frac{T^* \phi^* (1-\eta) * 860 * \alpha}{1.12 * 0.24 (t_1 - t_2)}$	5 회/hr
주 방	40~60 회/hr	

3-2. 풍량 선정

장비번호	실명	면적	천정고	체적	환기횟수	계산풍량	선정 풍량		비 고
		m ²	m	m ³	회/HR	CMH	수량	CMH	
<급기 FAN>									
SF-01	시험실 구역	2,542.2	3.0	7,627	3	22,880	1	22,940	IN LINE
	1F 전기적 기능 시험실	34.0	3.0	102	3	306	1	310	
	1F DRIVE 기능 시험실	48.0	3.0	144	3	432	1	440	
	1F DRIVE 시험실	333.0	3.0	999	3	2,997	1	3,000	
	1F 진동시험실	167.0	3.0	501	3	1,503	1	1,510	
	1F 극한 환경 시험실	33.0	3.0	99	3	297	1	300	
	1F 제어룸	23.0	3.0	69	3	207	1	210	
	1F 환경 시험실	261.6	3.0	785	3	2,354	1	2,360	
	1F EMI/EMS Shied실(추후) 등	217.6	3.0	653	3	1,958	1	1,960	
	1F 전기차/반도체/AZS IN-LINE검증시스템실	815.0	3.0	2,445	3	7,335	1	7,340	
	1F 대용량 모터부하실	257.0	3.0	771	3	2,313	1	2,320	
	1F 부하제어 및 기능시험실	41.0	3.0	123	3	369	1	370	
	1F 시험전원실	137.0	3.0	411	3	1,233	1	1,240	
	1F PLC/DRIVE 및 신사업 검증시스템	175.0	3.0	525	3	1,575	1	1,580	
SF-02	화장실 PIT	89.0	2.0	178	5	890	1	900	사류팬
<배기 FAN>									
EF-01	시험실 구역	2,542.2	3.0	7,627	3	22,880	1	22,940	IN LINE
	1F 전기적 기능 시험실	34.0	3.0	102	3	306	1	310	
	1F DRIVE 기능 시험실	48.0	3.0	144	3	432	1	440	
	1F DRIVE 시험실	333.0	3.0	999	3	2,997	1	3,000	
	1F 진동시험실	167.0	3.0	501	3	1,503	1	1,510	
	1F 극한 환경 시험실	33.0	3.0	99	3	297	1	300	
	1F 제어룸	23.0	3.0	69	3	207	1	210	
	1F 환경 시험실	261.6	3.0	785	3	2,354	1	2,360	
	1F EMI/EMS Shied실(추후) 등	217.6	3.0	653	3	1,958	1	1,960	
	1F 전기차/반도체/AZS IN-LINE검증시스템실	815.0	3.0	2,445	3	7,335	1	7,340	
	1F 대용량 모터부하실	257.0	3.0	771	3	2,313	1	2,320	
	1F 부하제어 및 기능시험실	41.0	3.0	123	3	369	1	370	
	1F 시험전원실	137.0	3.0	411	3	1,233	1	1,240	
	1F PLC/DRIVE 및 신사업 검증시스템	175.0	3.0	525	3	1,575	1	1,580	
<배기 FAN>									
TEF-01	화장실 배기	124.7	2.7	275	15	2,338	1	2,460	SIROCCO
	화장실 PIT	89.0	2.0	178	5	890	1	900	
	1F 장애인 화장실	7.6	2.7	21	15	309	1	400	
	1F 화장실	8.4	2.7	23	15	341	1	350	
	2F 화장실(남)	13.0	2.7	35	15	527	1	530	
	2F 화장실(여)	6.7	2.7	18	15	272	1	280	

3-3. FAN 선정

SF-01	
AIR VOLUME	
용 도	시험실 구역
면 적	2542.2 m ²
천 정 고	3.00 m
환기횟수	3.0 회/h
풍 량	22880 → 22940 m ³ /h
수 량	1 대

STATIC PRESSURE	
DESCRIPTION	pa
DIFFUSER/GRILLE/REGISTER	0
LOUVER	49
DUCT	
0.98 Pa/m × 100 m	98
0.78 Pa/m × m	-
PRE FILTER	
FITTINGS(DUCT 50%)	49
FLEXIBLE DUCT	9.8
RA PLENUM	
VOLUME DAMPER	29.4
FIRE DAMPER	9.8
SOUND ATTENUATOR	
PRE(RE) HEATING COIL	
GREASE FILTER	
VAV(CAV) UNIT	
B.D.D.	0
V.P	
SUB-TOTAL	245
SAFETY FACTOR(10%)	24.5
GRAND TOTAL	270

FAN SELECTION	
EQUIP. NO	SF-01
AIR VOLUME	22940 m ³ /h
MOTOR POWER	효율 : 40% 안전율 : 1.15
P = 22940 m ³ /h × 270 Pa ÷ (999.6 × 3600 × 0.4) × 1.15	
= 4.95 → 5.5 kW	
FAN SELECTION	
SPEC. : 22940 m ³ /h × 270 Pa × 5.5 kW	
Q'TY : 1 대	
TYPE : DUCT IN LINE	
MODEL Φ900	
LOCAT. 1F 전기차/반도체/AZS IN-LINE검증시스템실	
ELEC. : 3/380/60	
NOTE :	

SF-02	
AIR VOLUME	
용 도	화장실 PIT
면 적	89 m ²
천 정 고	2.00 m
환기횟수	5.0 회/h
풍 량	890 → 900 m ³ /h
수 량	1 대

STATIC PRESSURE	
DESCRIPTION	pa
DIFFUSER/GRILLE/REGISTER	29.4
LOUVER	49
DUCT	
0.98 Pa/m × 40 m	39.2
0.78 Pa/m × m	-
PRE FILTER	
FITTINGS(DUCT 50%)	19.6
FLEXIBLE DUCT	9.8
RA PLENUM	
VOLUME DAMPER	29.4
FIRE DAMPER	9.8
SOUND ATTENUATOR	
PRE(RE) HEATING COIL	
GREASE FILTER	
VAV(CAV) UNIT	
B.D.D.	0
V.P	
PRD DAMPER	0
SUB-TOTAL	186
SAFETY FACTOR(10%)	18.6
GRAND TOTAL	205

FAN SELECTION	
EQUIP. NO	SF-02
AIR VOLUME	900 m ³ /h
MOTOR POWER	효율 : 50% 안전율 : 1.2
P = 900 m ³ /h × 205 Pa ÷ (999.6 × 3600 × 0.5) × 1.15	
= 0.12 → 0.214 kW	
FAN SELECTION	
SPEC. : 900 m ³ /h × 205 Pa × 0.214 kW	
Q'TY : 1 대	
TYPE : DUCT IN LINE(사류팬)	
MODEL Φ350	
LOCAT. 1F 로비	
ELEC. : 1/220/60	
NOTE :	

EF-01

AIR VOLUME

용 도	시험실 구역		
면 적	2542.2 m²		
천 정 고	3.00 m		
환기횟수	3.0 회/h		
풍 량	22880	→	22940 m³/h
수 량	1 대		

AIR VOLUME

용 도			
면 적	m²		
천 정 고	m		
환기횟수	회/h		
풍 량		→	m³/h
수 량	대		

STATIC PRESSURE

DESCRIPTION	Pa
DIFFUSER/GRILLE/REGISTER	29.4
LOUVER	49
DUCT	
0.98 Pa/m × m	-
0.78 Pa/m × 100 m	78.4
FITTINGS(DUCT 50%)	39.2
FLEXIBLE DUCT	9.8
RA PLENUM	
VOLUME DAMPER	9.8
FIRE DAMPER	
SOUND ATTENUATOR	29.4
PRE(RE) HEATING COIL	9.8
GREASE FILTER	
VAV(CAV) UNIT	
B.D.D.	9.8
V.P	
SUB-TOTAL	265
SAFETY FACTOR(10%)	26.5
GRAND TOTAL	295

STATIC PRESSURE

DESCRIPTION	Pa
DIFFUSER/GRILLE/REGISTER	29.4
LOUVER	49
DUCT	
0.98 Pa/m × m	-
0.78 Pa/m × m	-
FITTINGS(DUCT 50%)	0
FLEXIBLE DUCT	
RA PLENUM	
VOLUME DAMPER	
FIRE DAMPER	0
SOUND ATTENUATOR	
PRE(RE) HEATING COIL	
GREASE FILTER	
VAV(CAV) UNIT	
B.D.D.	0
V.P	
PRD DAMPER	
SUB-TOTAL	78.4
SAFETY FACTOR(10%)	7.84
GRAND TOTAL	90

FAN SELECTION

EQUIP. NO	EF-01		
AIR VOLUME	22940	m³/h	
MOTOR POWER	효율 :	40%	안전율 : 1.2
P = 22940 m³/h×295 Pa÷(999.6×3600×0.4)×1.15			
= 5.41 → 5.5 kW			
FAN SELECTION			
SPEC. : 22940 m³/h × 295 Pa × 5.5 kW			
Q'TY : 1 대			
TYPE : DUCT IN LINE			
MODEL Ø900			
LOCAT. 1F 전기차/반도체/AZS IN-LINE검증시스템실			
ELEC. : 3/380/60			
NOTE :			

FAN SELECTION

EQUIP. NO			
AIR VOLUME	m³/h		
MOTOR POWER	효율 :	40%	안전율 : 1.2
0.00		→	0.4 kW
FAN SELECTION			
SPEC. :			
Q'TY :			
TYPE :			
MODEL :			
LOCAT. :			
ELEC. :			
NOTE :			

TEF-01

AIR VOLUME			
용 도	화장실 배기		
면 적	124.7 m ²		
천 정 고	2.70 m		
환기횟수	15.0 회/h		
풍 량	2337.5	→	2460 m ³ /h
수 량	1 대		

STATIC PRESSURE	
DESCRIPTION	Pa
DIFFUSER/GRILLE/REGISTER	29.4
LOUVER	49
DUCT	
0.98 Pa/m × m	-
0.78 Pa/m × 40 m	31.4
FITTINGS(DUCT 50%)	15.7
FLEXIBLE DUCT	9.8
RA PLENUM	
VOLUME DAMPER	29.4
FIRE DAMPER	9.8
SOUND ATTENUATOR	
PRE(RE) HEATING COIL	
GREASE FILTER	
VAV(CAV) UNIT	
B.D.D.	9.8
V.P	
SUB-TOTAL	184
SAFETY FACTOR(10%)	18.4
GRAND TOTAL	205

FAN SELECTION	
EQUIP. NO	TEF-01
AIR VOLUME	2460 m ³ /h
MOTOR POWER	효율 : 40% 안전율 : 1.15
$P = 2460 \text{ m}^3/\text{h} \times 205 \text{ Pa} \div (999.6 \times 3600 \times 0.4) \times 1.15$ $= 0.41 \rightarrow 0.400 \text{ kW}$	
FAN SELECTION	
SPEC. : 2460 m ³ /h × 205 Pa × 0.4 kW	
Q'TY : 1 대	
TYPE : SIROCCO	
MODEL # 2.0 SS	
LOCAT. 지붕층	
ELEC. : 1/220/60	
NOTE :	

AIR VOLUME	
용 도	
면 적	m ²
천 정 고	m
환기횟수	회/h
풍 량	→ m ³ /h
수 량	대

STATIC PRESSURE	
DESCRIPTION	Pa
DIFFUSER/GRILLE/REGISTER	29.4
LOUVER	49
DUCT	
0.98 Pa/m × m	-
0.78 Pa/m × m	-
FITTINGS(DUCT 50%)	0
FLEXIBLE DUCT	
RA PLENUM	
VOLUME DAMPER	9.8
FIRE DAMPER	9.8
SOUND ATTENUATOR	
PRE(RE) HEATING COIL	
GREASE FILTER	
VAV(CAV) UNIT	
B.D.D.	0
V.P	
SUB-TOTAL	98
SAFETY FACTOR(10%)	9.8
GRAND TOTAL	110

FAN SELECTION	
EQUIP. NO	
AIR VOLUME	m ³ /h
MOTOR POWER	효율 : 안전율 : 1.15
0.00 → 1.5 kW	
FAN SELECTION	
SPEC. :	
Q'TY :	
TYPE :	
MODEL :	
LOCAT. :	
ELEC. :	
NOTE :	

3-4. 전열교환기 선정

LS ELECTRIC(천안사업장) R-Center 구축

RM No.	실 명	수량	면적	인원	단위 풍량	계산풍량	적용풍량	장비선정
			m²	인	CMH	CMH	CMH	CMH
환기유니트 풍량 선정		1	762.8	200	－	5,800	6,250	6,250
부대시설								
1002	1F_ 로비	1	154.3	17	29	493	500	500
1024	1F_ 고객 대기실-1	1	14.1	8	29	232	250	
1025	1F_ 고객 대기실-2	1	14.1	8	29	232	250	
1026	1F_ 고객 대기실-3	1	14.1	8	29	232	250	800
1027	1F_고객접견실	1	45.8	12	29	348	550	
2001	2F_ Software Lab(20명)	1	67.2	21	29	609	650	800
2010	2F_복도	1	44.1	5	29	145	150	
2007	2F_자료실	1	7.8	1	29	29	50	250
2008	2F_검증시스템 Control Room	1	27.8	6	29	174	200	
2009	2F_세미나실/다목적실(32석)	1	77.6	35	29	1,015	1,000	1,000
2010	2F_휴게공간	1	44.1	9	29	261	200	800
2011	2F_대회의실	1	30.8	16	29	464	500	
2012	2F_OA/탕비실	1	15.8	3	29	87	100	
2013	2F_사무실 (30석)	1	167.7	34	29	986	1,000	
2014	2F_중회의실-1	1	11.8	7	29	203	250	800x2대
2015	2F_중회의실-2	1	11.9	7	29	203	250	
2016	2F_센터장실	1	13.8	3	29	87	100	

제 4장 위생설비

4-1. 배수 설비

4-1. 배수 설비

1) PIT 내 집수정 배수 Pump 선정 SP-1

① 유량(V) : 집수정을 7분만에 배수할 수 있는 유량으로 선정한다.

Sump Size : 1,000L × 1,000W × 1,000H 기준하여 150lpm 선정한다

② 양정(H)

- 건물압상고 :	3.7m
- 토출 속도 수두 :	2.0m
- 밸브 손실 수두 :	25 kPa
- 배관손실 : 10m × 0.49kPa/m × 1.5 =	8 kPa
- Safety(10%)	7 kPa
	96 kPa → 100kPa

③ 선정

- 형 식 : 수중 Pump

- $\varnothing 50 \times 150\text{lpm} \times 100\text{kPa} \times 1.5\text{kW}$

- 수 량 : 2대 (1대 예비)

- 설치위치 : PIT층